

CHAPTER 5

Noah Parker

5.1. Let $f : A \rightarrow B$ be an integral homomorphism of rings. Show that $f^* : \text{Spec } B \rightarrow \text{Spec } A$ is a closed mapping. (This is a geometric equivalent of (5.10).)

Proof. Take $\mathfrak{b} \trianglelefteq B$, and recall that $f^*(V(\mathfrak{b})) \subset \overline{f^*(V(\mathfrak{b}))} = V(\mathfrak{b}^c)$ by (Ex. 1.21iii). Now, suppose $\mathfrak{p} \in V(\mathfrak{b}^c)$ is a prime ideal of A , i.e. $\mathfrak{p} \supset \mathfrak{b}^c = f^{-1}(\mathfrak{b})$. Then in particular $\mathfrak{p} \supset \ker f$, so $\mathfrak{p}$ descends to a prime ideal $f(\mathfrak{p})$ of $A/\ker f \cong \text{im } f \subset B$.

$\text{im } f \subset B$ is integral, so (5.10) gives us a prime $\mathfrak{q} \in \text{Spec } B$ with $\mathfrak{q} \cap \text{im } f = f(\mathfrak{p})$. We have that $\mathfrak{q} \supset f(\mathfrak{p}) \supset f(\mathfrak{b}^c)$ implies $\mathfrak{q} \supset \mathfrak{b}^{ce} = \mathfrak{b}$, i.e. $\mathfrak{q} \in V(\mathfrak{b})$. Further,

$$\begin{aligned} f^*(\mathfrak{q}) &= f^{-1}(\mathfrak{q}) \\ &= f^{-1}(\mathfrak{q} \cap \text{im } f) \\ &= f^{-1}(f(\mathfrak{p})) \\ &= \mathfrak{p}, \end{aligned}$$

since $\mathfrak{p} \supset \ker f$. Thus, $\mathfrak{p} \in f^*(V(\mathfrak{b}))$, so in fact $f^*(V(\mathfrak{b})) = \overline{f^*(V(\mathfrak{b}))}$ is closed. $\square$

5.2. Let B be integral over $A \subset B$, and let $f : A \rightarrow \Omega$ be a homomorphism of A into an algebraically closed field Ω . Show that f can be extended to a homomorphism of B into Ω .

Proof. $f : A \rightarrow \Omega$ corresponds to a maximal ideal $\mathfrak{p} = \ker f \trianglelefteq A$ and an inclusion $\bar{f} : A/\mathfrak{p} \hookrightarrow \Omega$. (5.10) supplies us with a prime ideal $\mathfrak{q} \trianglelefteq B$ such that $\mathfrak{q} \cap A = \mathfrak{p}$, which is in fact maximal by (5.8). Then $\pi : B \rightarrow B/\mathfrak{q}$ restricts to a map

$$\pi|_A : A \rightarrow \frac{A}{\mathfrak{q} \cap A} = \frac{A}{\mathfrak{p}} \subset \frac{B}{\mathfrak{q}}.$$

(5.6) gives us that $K := B/\mathfrak{q}$ is integral over $k := A/\mathfrak{p}$.

Let $(k, \bar{f}) \leq (C, \bar{g})$ be a maximal element of Σ . In particular, it is a valuation ring of K , so $K \supset C \supset \bar{k} = K$ by (5.22), whence $K = C$. Then $g := \bar{g} \circ \pi : B \rightarrow \Omega$ has

$$g|_A = \bar{g}|_k \circ \pi|_A = \bar{f} \circ \pi|_A = f.$$

 $\square$

5.10. A ring homomorphism $f : A \rightarrow B$ is said to have the *going-up property* (resp. the *going-down property*) if the conclusion of the going-up theorem (resp. going-down theorem) holds for B and its subring $f(A)$. Let $f^* : \text{Spec } B \rightarrow \text{Spec } A$ be the mapping associated with f .

(a) Consider the following three statements:

- (i) f^* is a closed mapping.
- (ii) f has the going-up property.
- (iii) For any prime ideal $\mathfrak{q}$ of B , let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \text{Spec}(B/\mathfrak{q}) \rightarrow \text{Spec}(A/\mathfrak{p})$ is surjective.

Prove that (i) $\implies$ (ii) $\iff$ (iii).

(b) Consider the following three statements:

- (i) f^* is an open mapping.
- (ii) f has the going-down property.
- (iii) For any prime ideal $\mathfrak{q}$ of B , let $\mathfrak{p} = \mathfrak{q}^c$. Then $f^* : \text{Spec}(B_{\mathfrak{q}}) \rightarrow \text{Spec}(A_{\mathfrak{p}})$ is surjective.

Prove that $(i) \implies (ii) \iff (iii)$.

- 5.16. Let k be a field and let $A \neq 0$ be a finitely generated k -algebra. Then there exist elements $y_1, \dots, y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \dots, y_r]$.

We shall assume that k is *infinite*. (A different proof is required for the finite case.) From the proof, it follows that $y_1, \dots, y_r$ may be chosen to be linear combinations of $x_1, \dots, x_n$. This has the following geometrical interpretation: if k is algebraically closed and X is an affine algebraic variety in k^n with coordinate ring $A \neq 0$, then there exists a linear subspace of dimension r in k^n and a linear mapping of k^n onto L which maps X onto L . [Use Exercise (2).]